

SYNOPSIS: AI-Optimized Post-Quantum SCTP Association Gateway

1. Title

Autonomous Post-Quantum Secure Gateway with AI-Driven Adaptive Cryptography and SCTP Multi-Homing Resiliency

Post-Quantum Secure SCTP Association Gateway is designed to bridge the gap between the classical transport protocols and the advanced cryptographic requirements of the "Post-Quantum" era mentioned in 2026 research trends. This project transforms a standard legacy connection into a high-resiliency, quantum-resistant communication channel.

Project Overview & Core Objectives

The objective is to build a middleware gateway that intercepts standard TCP or UDP traffic and tunnels it through a Stream Control Transmission Protocol (SCTP) association. This association is secured using a hybrid cryptographic handshake that combines classical algorithms (like RSA or Diffie-Hellman) with Post-Quantum Cryptography (PQC).

- Target Research Gap: Addresses the "Harvest Now, Decrypt Later" threat, where attackers capture current traffic to decrypt it once quantum computers become viable. It also solves the "SCTP Transition" gap, as SCTP's superior multihoming features are often underutilized due to legacy middlebox incompatibility.

Innovative Components (AI & Cybersecurity)

A. AI-Driven Adaptive Cryptography [Innovative Component]

The gateway does not use a static encryption level. Instead, it features an AI Intelligence Module (trained using Reinforcement Learning or Random Forest models) that monitors the network environment in real-time.

- Mechanism: The AI analyzes network "jitter," packet loss, and device energy levels (critical for IoT nodes) to dynamically select PQC parameters.
- Benefit: If the device is low on battery or the network is stable, the AI selects a more computationally efficient lattice-based algorithm (like ML-KEM/Kyber-512). If a high-threat "anomaly" is detected via traffic analysis, the AI scales up to the most robust 2026 standards (Kyber-1024).

B. Hybrid Post-Quantum Handshake

The project modifies the standard SCTP four-way handshake ($INIT \rightarrow INIT\text{-}ACK \rightarrow COOKIE\text{-}ECHO \rightarrow COOKIE\text{-}ACK$).

- Security Logic: The $INIT$ chunk is modified to include a "PQC Key Encapsulation" payload alongside the traditional Diffie-Hellman parameters. This ensures that even if a quantum computer breaks the classical $g^{xy} \pmod p$ exchange in the future, the session remains secured by the quantum-resistant layer.

C. AI-Enhanced Multihoming Path Selection

SCTP allows an association to span multiple IP addresses (Multihoming).

- AI Integration: A supervised learning model monitors the performance of the "primary" and "backup" paths. If the AI detects patterns indicative of a Man-in-the-Middle (MitM) attack or a "Sinkhole" on the primary path, it autonomously triggers a "Changeover" to the secondary path before the connection drops.

Technical Implementation Details

Component	Technology/Protocol	Role
Transport	SCTP (libsctp)	Handles multi-streaming and multihoming redundancy.
Classical Crypto	RSA / ECC (Unit IV)	Provides baseline authentication and compatibility [Image 1].
PQC Layer	ML-KEM (Kyber)	Provides quantum-resistance against Shor's algorithm.
AI Engine	Python (Scikit-Learn)	Performs real-time anomaly scoring and adaptive scaling.
Network Interface	Berkeley Sockets (C)	Uses <code>socket()</code> , <code>bind()</code> , and <code>inet_pton</code> for associations.

Implementation Roadmap for Students

- Socket Layer:** Develop a C-based gateway using `SOCK_SEQPACKET` to handle SCTP associations. Use `sctp_bindx()` to bind multiple local IP addresses for multihoming tests.
- Cryptographic Wrapper:** Integrate the OQS (Open Quantum Safe) library to handle the PQC key exchange during the association startup.
- AI Integration:** Export traffic features (packet size, inter-arrival time) to a local Python script via a Unix Domain Socket. The Python script returns a "Security Policy" (e.g., "High-Security Mode" or "Power-Saving Mode").
- Testing:** Simulate a quantum-adversary node using a network emulator (GNS3 or NS3) and verify that the gateway can maintain a secure session even when classical keys are "intercepted".

2. Abstract

With the advent of the "Post-Quantum" era, traditional cryptographic standards such as RSA and Diffie-Hellman are increasingly vulnerable to Shor's algorithm. Additionally, legacy transport protocols (TCP/UDP) lack the multi-homing resiliency needed for the decentralized infrastructure of 2026. This project proposes an **SCTP Association Gateway** that intercepts legacy application traffic and tunnels it through a quantum-resistant channel. The system utilizes **Hybrid Cryptography**, combining classical RSA with **Lattice-based PQC (ML-KEM/Kyber)** to mitigate "Harvest Now, Decrypt Later" risks. An integrated **AI Intelligence Module** monitors network performance and device energy levels to dynamically scale encryption strength. By leveraging the **Stream Control Transmission Protocol (SCTP)**, the gateway ensures high availability through AI-managed autonomous path failovers.

3. Problem Statement

Existing Network Infrastructures:

- **Quantum Vulnerability:** Standard 3-way handshakes are decryptable by future quantum computers.
- **Path Fragility:** TCP/UDP connections fail during IP changes or interface drops, causing session loss.
- **Static Security Overhead:** Current encryption does not adapt to the energy constraints of IoT nodes, leading to rapid battery depletion.

There is a need for a system that ensures:

- Quantum-resistant data integrity and confidentiality.
- Fault-tolerant transport using multi-streaming and multi-homing.
- AI-driven optimization between security depth and resource consumption.

4. Objectives

- To design a C-based gateway using **Berkeley Sockets** and **libsctp**.
- To implement a **Hybrid PQC Handshake** (Classical RSA + NIST-approved Kyber).
- To develop an **AI Intelligence Module** for adaptive security scaling based on network jitter and battery life.
- To create an **AI-Enhanced Path Selection** mechanism for autonomous SCTP changeovers.
- To ensure transparent compatibility with legacy applications through local socket proxying.

5. Methodology

- **Step 1: Traffic Interception:** Setup a listening socket using `socket()`, `bind()`, and `listen()` to intercept legacy traffic.
- **Step 2: AI Environment Sensing:** Use a Python-based agent to extract real-time features (packet loss, throughput, device battery %).
- **Step 3: Secure SCTP Handshake:** Initiate a 4-way association. Inject PQC Key Encapsulation (KEM) payloads into the `INIT` and `INIT-ACK` chunks.

- **Step 4: Encrypted Tunneling:** Derive session keys and encrypt data using AES-256-GCM.
- **Step 5: Autonomous Multi-homing:** Monitor primary and secondary paths. If the AI detects a "Sinkhole" or high-latency pattern, it triggers an autonomous `sctp_bindx()` changeover.

6. Tools & Technologies

- **Programming:** C (Gateway Logic), Python (AI/ML Engine).
- **Transport:** SCTP (libsctp), `SOCK_SEQPACKET`.
- **Cryptography:** Open Quantum Safe (OQS) Library, ML-KEM (Kyber).
- **AI/ML:** Scikit-learn (Random Forest for anomaly detection and mode selection).
- **Platform:** Linux (Ubuntu 24.04/26.04).

7. Expected Outcomes

- A communication channel resistant to future quantum decryption attempts.
- Uninterrupted session availability during network interface failures (zero-packet-loss failover).
- Dynamic resource management resulting in increased battery longevity for IoT nodes.
- Verification of **Course Outcome 4 (CO4)** by securing data across wired and wireless SCTP paths [Image 2].

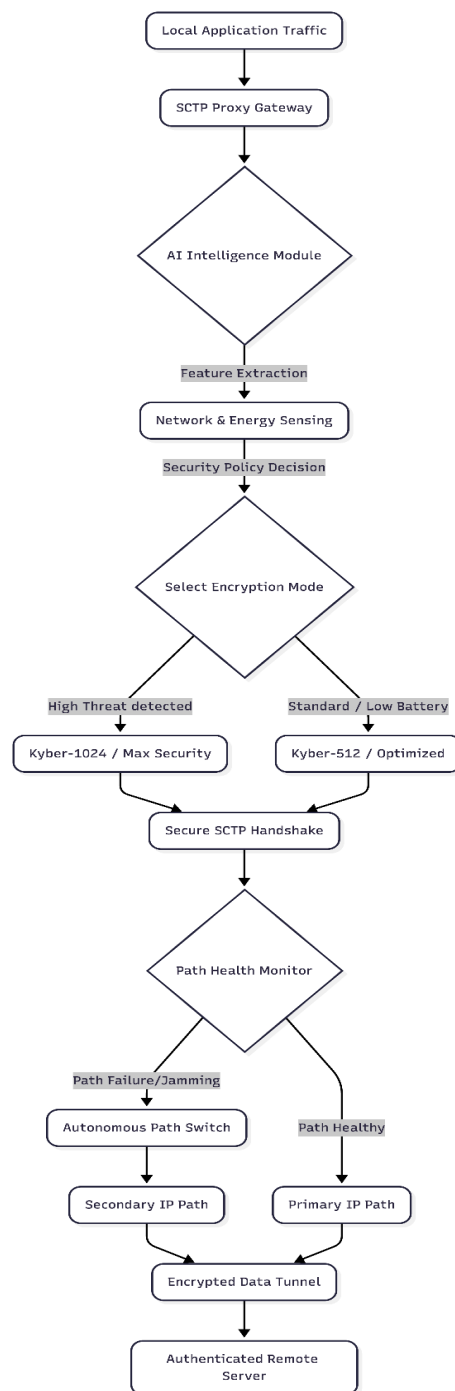
8. Applications

- **Military Communications:** Resilient, multi-path tactical links for battlefield networks.
- **Critical Infrastructure:** Securing Smart Grid WSNs against state-sponsored quantum threats.
- **Banking:** Protecting long-term archival data and high-frequency transactions from future decryption.

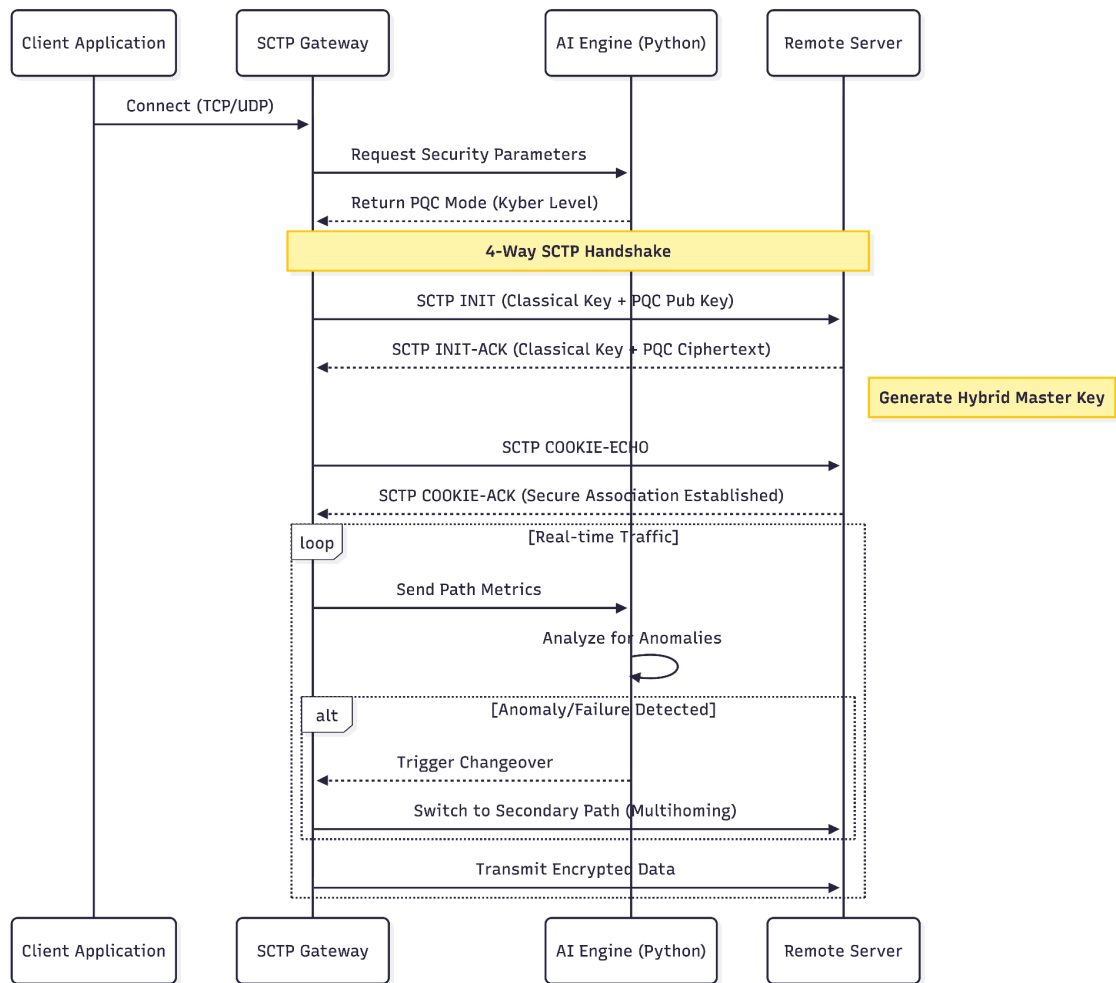
9. Conclusion

This project bridges the gap between Unit I (Transport Layer) and Unit IV (Cryptography) of the syllabus while solving the 2026 research gap in **Post-Quantum IoT Readiness**. By merging AI-driven adaptation with the robust multi-homing of SCTP, the gateway provides a future-proof architecture for secure digital highways.

MAIN SYSTEM FLOW



HYBRID PQC HANDSHAKE + AI ANALYSIS FLOW



NETWORK TOPOLOGY (MULTI-HOMED ARCHITECTURE)
 (This visualization illustrates the "Path Selection" research gap)

Component	Interface 1 (Primary)	Interface 2 (Secondary)
Gateway Link	192.168.1.10 (Wired)	10.0.5.20 (LTE/5G)
AI Monitor	High Latency Detection	Path Entropy Analysis
PQC Layer	ML-KEM Resistance	AES-256 Integrity Check